

Mathieu Waharte

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Profile

Software Engineer, specializing in Full-Stack Development and Artificial Intelligence.

Education

Polytech Paris-Saclay (Université Paris-Saclay) <i>Diplôme d'Ingénieur (M.Sc.) in Computer Science & Applied Mathematics</i> - Advanced C++, HPC, Distributed & Graph Algorithms, ILP, SVD, ODE & PDE solving, NLP, Cybersecurity	Orsay, France 2023 – Present
Paris-Saclay University <i>Bachelor of Science (B.S.) in Computer Science - with high honors</i> - Supervised Learning, Object-Oriented and Functional Programming, Logic, Computability	Orsay, France 2021 – 2023
L'Essouriau High School, Paris-Saclay University <i>Intensive Preparatory Class in Math and Physics for Engineering School (CPGE)</i> - Mathematics, Physics, Modeling, Problem Solving	Les Ulis, France 2019 – 2021

Experience

Fullstack Engineer Apprentice <i>Dassault Systèmes</i> - Full-stack design and development for the Scientific Notebook and Reaction Planner applications in an Agile/Scrum environment (Angular, Java, TypeScript, Python, SPARQL) with maintainable documentation. - Led the development (using design patterns) of a distributed computing queue, from specification and PoC to production, coordinating with Dev/PM/QA/UX teams, resulting in an increase from 40 to 100% reliability, even under load . - Engineered a robust LangChain AI agent and an MCP server to convert molecules into a standardized format, compute synthesizability scores, and interpret them from natural-language prompts through dedicated tools. Validated the system with a mock MCP server, PACT flow contracts, and tests evaluating LLM routing decisions . All errors are handled through custom states and pre/post-processing. - Autonomously delivered features across multiple environments (Java, Node.js, Docker, Python, Angular), ensuring quality through unit/integration testing (Jasmine, Page Objects, JMeter) and code reviews. Provided rigorous training and documentation for team use.	2023 – Present Vélizy-Villacoublay, France
Research Intern (Natural Language Processing - NLP) <i>Department of Artificial Intelligence, Faculty of ICT, University of Malta</i> - Conducted a literature review on Text-to-Speech (TTS). Developed a paper search application featuring a ranking and filtering system via Streamlit , enabling the analysis of several thousand articles in just a few hours. - Processed audio data (10h), performed fine-tuning and inference on the XTTS model using Google Colab (PyTorch & GPT-2) to integrate Maltese without regression. Robust and documented code is available on GitHub and Hugging Face.	June 2025 – August 2025 Malta

Projects

Band aware dual perspective ResNet for Escampe <i>Negamax, ResNet, PyTorch, Java, Machine Learning</i> - Engineered BandDPER , a custom 730K-parameter Siamese ResNet evaluation model featuring dual-perspective CNN encoders, piece-relative spatial mapping (HalfKP), and a direct output bypass for forced-pass detection. - Implemented adversarial search via Negamax with Alpha-Beta pruning , optimized with Iterative Deepening , Killer Heuristic, and History Heuristic to achieve aggressive pruning. - Bootstrapped a dataset of 3 million positions (6M mirrored) using parallel depth-7 minimax search, and trained the model using AdamW and Cosine Annealing with dropout regularization.	March 2026
Hackathon Deloitte x Google Cloud- 1st Place Winner <i>Google Cloud, Agent Development Kit</i> - Developed a distributed multi-agent system using ADK and the MCP protocol to automate marketing campaign creation. Orchestrated specialized agents communicating through the A2A protocol. - Integrated cloud-native services to perform demographic queries via BigQuery, generate multimodal content with Gemini and Imagen, and deploy on Cloud Run and Vertex AI Agent Engine.	November 2025
Dynamic Simulation - Magnus Carlos <i>SDL2, C++, Physical Modeling</i> - Managed Agile development of a scientific simulation and its 3D rendering for the Magnus effect with a team of 6 students. - Performed numerical solving and implemented the system in C++ (real-time visualization with SDL2) to reproduce, model, and dynamically predict the behavior of a complex physical system (trajectories, fluids).	2024

Technical Skills

Languages : JavaScript/TypeScript (Angular, Node.js, Express, React, Vue 3), C++, Java, Python (FastAPI), C, SQL, Bash
Quality & Software Testing : PACT, Jasmine, Jest, Karma, Page Objects, PCS (JMeter), JUnit, Mocks, TDD
Artificial Intelligence & Data : LLM, RAG, LoRA, LangChain, MCP, PyTorch, NumPy, pandas, OpenCV, scikit-learn
Machine Learning : AI agents, model training and evaluation, data preprocessing, Big Data, PCA, un/supervised learning
Simulation & Numerical Computation : ODE/PDE solving, probabilistic models, MPI, OpenMP, CUDA
Design, Methods & Databases : API REST, Design Patterns, UML, Postgres, MySQL, RDF, SQLite, Prisma, Redis
Tools and DevOps : Git, CI/CD, Docker, Postman, OpenAPI, Packaging, Tomcat, VMware, Hugging Face, LaTeX

Certificates and Languages

MOOC SecNumAc in cybersecurity - ANSSI
Model Context Protocol : Advanced Topics - Anthropic
Label Handimanagement - Companieros & Dassault Systèmes
English : C1 (TOEIC, 990), full professional proficiency